

Fictional case study for training. Not a real investigation.



Investigation report

Rejected take-off and runway overrun, Tarlton Springs - AAIB-style
causal and contributory factors report

Drafted 05/10/2026

INVESTIGATION WORKFLOW SUITE

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Synopsis

On 18/06/2026 Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, was departing runway 09 at Tarlton Springs Aerodrome after last light with 44 passengers, 2 flight crew and 1 cabin crew. Two seconds after V1 the first officer saw two kangaroos moving onto the runway from the northern side. The pilot in command rejected the take-off at 118 kt, 6 kt above V1. About one second later the right main landing gear and the No. 2 propeller struck both animals, which left reverse thrust on the No. 1 engine only. The aircraft left the runway end at about 21 kt. It came to rest upright on the extended centreline, 62 m into the 90 m runway end safety area and 28 m short of the end of the prepared area. There were no serious injuries, although one cabin crew member reported bruising during the disembarkation. The aircraft was substantially damaged. Three No. 2 propeller blades were beyond limits and the engine required a shock-load inspection. The right main gear doors were distorted, both right main tyres deflated through their fuse plugs, and the lower right fuselage was punctured. Both kangaroos were killed and the aerodrome was closed for 4 hours 20 minutes.

The overrun had two causes, and both were needed to consume the scheduled 67 m margin. The stop began 6 kt late. Deceleration was also approximately 15% below the scheduled accelerate-stop performance because the strike removed the No. 2 reverse. The kangaroos very likely entered through a section of the northern perimeter fence whose repair had been deferred, unfunded, for three budget cycles. The aerodrome's sighting log showed the hour after last light to be the peak of kangaroo movement, and the departure's timing in that hour likely affected it. No runway inspection was requested, which likely contributed. The operator required one only at aerodromes listed after a reported wildlife strike. That criterion very likely left the control unarmed at Tarlton Springs despite eleven logged sightings of kangaroos on the sealed surface. The evening service had likely been introduced without reopening a route risk assessment that restricted the aerodrome to daylight operations. The aerodrome standard's size threshold exempted Tarlton Springs from documenting a wildlife plan. It was found likely to have contributed, though with less confidence than the others. Rejected take-off training covered only aircraft system failures, and this likely contributed to the decision to reject above V1. However, the pilot in command knew the rule and departed from it deliberately.

Two matters increased risk without contributing. The cabin floor proximity lighting was partially unserviceable through a defect that routine checks could not detect. The crew were also on the fifth sector of a duty that the schedule routinely extends towards its limit. An engine power loss and a tailwind were ruled out. Whether the runway edge lighting intensity impaired detection of the animals could not be determined. The recovery was well executed, and maximum braking was established faster than the fleet mean, which limited the overrun.

The operator suspended the evening service and on 26/06/2026 amended Appendix 8A to list Tarlton Springs. The aerodrome operator installed a temporary electric wire and funded the fence replacement for 09/2026. A recommendation was made to the operator to list aerodromes on wildlife exposure rather than on a reported strike. It also asks the operator to add a route risk assessment trigger for any aerodrome first operated into after last light.

Factual information

History of the flight

On 18/06/2026 Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, was the last scheduled sector of the day from Tarlton Springs Aerodrome. It was the fifth sector of the flight crew's duty. The sector had been added to their duty that morning, when the rostered crew reported unfit. There were 44 passengers, 2 flight crew and 1 cabin crew on board.

The aerodrome operator had made its only runway inspection of the day at 21:35 UTC on 17/06/2026 (07:35 local on 18/06/2026), before the first scheduled arrival, and had recorded the movement area as clear. No further inspection was carried out or requested.

The doors closed at 07:16 UTC (17:16 local), 40 minutes behind schedule, with the aerodrome in darkness. The first officer was pilot flying. V1 had been calculated as 112 kt for 20,800 kg, flap 10, a dry runway 09 and a 5 kt headwind component. At 07:29:10 the tower cleared SB512 for take-off on runway 09 and passed the wind as 090 at 6 kt. The tower did not offer a runway inspection, and the crew did not request one. No wildlife transmission had been made on the tower channel in the preceding 90 minutes.

The take-off roll began at 07:30:02, and the pilot in command called V1 at 07:30:29. Two seconds later the first officer called "roos, roos, on the left". Two kangaroos were moving onto the runway from the northern side, past the 1,100 m point. She later estimated that they were about 200 m away when she first saw them, although she said the distance could have been less. At 07:30:32 the pilot in command called "Stopping. I have control." He rejected the take-off at 118 kt, which was 6 kt and 3 seconds after V1. At that point the aircraft was 1,152 m along runway 09, with 695 m of sealed surface remaining. About one second later the right main landing gear and the No. 2 propeller struck both animals.

Maximum braking was applied, but reverse was obtained on the No. 1 engine only. At 07:30:47 the pilot in command said "We're not stopping on it." The aircraft left the end of runway 09 at about 21 kt at 07:30:51. It came to rest at 07:30:57, 62 m into the 90 m runway end safety area (RESA). It was upright on the extended centreline, with the nose wheel in graded gravel, 28 m short of the end of the prepared area.

The pilot in command set the parking brake and found no fire or abnormal indications, so he decided not to evacuate. Both engines were shut down at 07:31:26. At 07:31:34 he told the cabin to remain seated. At 07:31:54 he advised the tower and asked it to activate the aerodrome emergency plan. The tower did so at 07:32:10 and dispatched a vehicle at the same time. The forward airstair was recorded operating at 07:34:40. Passengers disembarked by the airstair without slides, and the main cabin lights stayed on throughout. There was no fire.

The occurrence notification states that passengers were disembarked 4 minutes 8 seconds after the aircraft stopped. The sequence of events, however, gives disembarkation as running from 07:34:40 to 07:38:48. That is a period of 4 minutes 8 seconds, but it began after the airstair started operating.

No serious injuries were sustained. One cabin crew member reported bruising during disembarkation. Both kangaroos were killed, and the aerodrome was closed for 4 hours 20 minutes. Both flight crew were drug and alcohol tested, with negative results. The Head of Flying Operations suspended the evening Tarlton Springs service.

Recorded information

The operator's engineering representative removed the FDR and CVR at 12:40 UTC on 18/06/2026. The operator's flight data section then issued its FDR and QAR readout as report FDA-2026-0417-R1 on 24/06/2026.

Until the power levers were retarded at 07:30:32, both engines were at take-off torque, at 98.4% and 98.5%, with no engine exceedance, caution or warning. Fuel flow, ITT and propeller RPM were within normal scatter. The recorded headwind component was 4 to 6 kt throughout the roll, and there was no tailwind component.

Both power levers reached flight idle within 0.6 seconds of the reject. At the impact at 07:30:33 the data showed a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. Maximum braking at 2,940 psi was established at 07:30:34. This was 2 seconds after the decision, against the operator's recorded fleet mean of 3.2 seconds. Anti-skid was active on all four wheels. Reverse was selected at 07:30:36, at 98 kt, but on No. 1 only. No. 2 propeller RPM decayed from 07:30:33, and the No. 2 reverse interlock did not make.

Mean deceleration between 07:30:34 and 07:30:44 was 2.6 m/s². That is approximately 15% below the certified accelerate-stop value for that weight and runway condition, a value which assumes symmetric reverse. In the gravel, deceleration fell to 1.9 m/s². Reconstruction from groundspeed gave a stopping distance of 757 m from 118 kt.

After the aircraft stopped, the CVR captured the first officer saying that the animals "came out of the dark on the left of the strip, crossed the centreline". She said she had them in view for perhaps two seconds or less. The pilot in command said "I never had them". He added that he had been looking at the engine page after V1.

Aircraft information

[NOT ESTABLISHED: aircraft type, maintenance history and serviceability before the occurrence.]

Three No. 2 propeller blades were damaged beyond repairable limits. They showed leading edge deformation, and one blade tip had separated. Both right main tyres had deflated through the fuse plugs, consistent with brake energy. The right main gear doors were distorted, although the gear structure was within limits. The lower right fuselage skin showed abrasion and two punctures consistent with propeller debris.

During disembarkation the emergency lighting was armed and the ceiling lights were on. The first officer noticed unlit sections of the aisle floor proximity strip towards the rear of the cabin. She reported them to the engineer the following morning.

Meteorology

The headwind component recorded during the roll was consistent with the 07:00 UTC routine observation. Runway 09 was dry. [NOT ESTABLISHED: the time of last light and the end of civil twilight on 18/06/2026.]

Personnel information

The pilot in command had 8,410 hours in total, 2,140 of them on type, and had been in command for 3 years 1 month. His last cyclic simulator check, on 11/03/2026, and his last line check, on 02/12/2025, were both satisfactory. He had no prior occurrence history. He reported that he had seen kangaroos on the strip at Tarlton Springs in the evening four or five times, but he had never reported a sighting.

The first officer had 1,655 hours in total, 405 of them on type. Her last cyclic simulator check, on 28/04/2026, was satisfactory. Her two previous sectors into Tarlton Springs, on 09/01/2026 and

17/02/2026, were both flown in daylight. Before the flight she reviewed the flight plan, the charts, the ERSA entry and the company aerodrome brief page. She reported that none of them mentioned wildlife.

The duty record shows a sign-on at 06:30 local and a planned duty of 12 hours 5 minutes, against a maximum of 12 hours 30 minutes for five sectors. The occurrence took place 11 hours 0 minutes into that duty. Both pilots' duty and rest in the preceding 7 days were within company limits. Both described themselves as tired but not struggling. The pilot in command said he would have declined a further sector.

Aerodrome information

Beyond the sealed surface of runway 09 lies a 90 m RESA of graded gravel, and beyond that the terrain is ungraded. The tower was rostered to close at 18:00 local.

The survey recovered the two kangaroo carcasses 1,188 m and 1,221 m from the runway 09 threshold. Continuous tyre marks ran from 1,160 m to the runway end, and the nose wheel rut continued to 62 m beyond it.

The two pilots described the runway lighting differently. The pilot in command found it brighter than he would want, with darkness on either side. He could not tell, however, whether this was why he had not seen the animals. The first officer found the lighting normal, though she had only two daylight landings there for comparison. She said she could see the animals clearly once they were on the sealed surface.

[NOT ESTABLISHED: condition and repair history of the aerodrome perimeter fence.]

Organisational and management information

The operator's Part A section 8.14 refers to an Appendix 8A list of aerodromes at which a runway inspection is requested before the first departure after dark. Tarlton Springs was not on that list. [NOT ESTABLISHED: the criterion for inclusion on Appendix 8A and the content of the Safety Department's review of 03/03/2025.]

According to the pilot in command, the evening service began around the middle of the previous year, and before that Tarlton Springs had been a daylight operation. He said that no briefing or change accompanying the evening service reached him.

Both pilots stated that their rejected take-off training had always involved engine or configuration events below V1, and never an obstruction on the runway. The standard take-off brief gives the captain the decision to reject below V1. Above V1, it requires the crew to continue.

Tests and research

The engineering inspection took place from 19/06/2026 to 21/06/2026. Animal tissue was recovered from two No. 2 propeller blade roots. The engine was subjected to a shock-load inspection. Borescope inspection of both engines found no pre-existing damage and no evidence of any power anomaly before the impact. Brake energy absorbed was calculated at 84% of the certified maximum. There was no brake fire or hydraulic leak, and the anti-skid system tested serviceable.

A functional check of the cabin floor proximity strip was carried out on 24/06/2026. [NOT ESTABLISHED: result of the check.]

Other information

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The aerodrome kept a paper wildlife sighting log in its office, covering 01/2025 to 06/2026. Of its 46 entries, 41 recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was not reported to any operator. The aerodrome reporting officer confirmed that no airline had ever asked for it.

Analysis

External Influences

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Aerodrome Standard AS-139 treats the duty to manage wildlife and the duty to document that management as separate duties. Section 10.4.1 requires the operator of every certified aerodrome to "take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome." A documented plan, however, is required only where section 10.4.2 applies. That is where the aerodrome "is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year." Tarlton Springs met neither limb. Its heaviest scheduled aeroplane was the 22,900 kg twin-turboprop used on the Saltbush Air service, and it handled 61,400 passenger movements in 2025. Under 10.4.3, the measures it took therefore "need not be documented."

Both the certified aerodrome register and the aerodrome operator's own compliance file confirmed this position, and the aerodrome was not in breach of the standard. It is virtually certain that this condition existed. The matter is analysed here as an external influence rather than as a question of compliance. The reason is that the threshold in the standard itself left an aerodrome of this size without a documented wildlife programme.

In practice, the aerodrome held no wildlife hazard management plan, no hazard assessment and no dispersal programme. Its only documented wildlife measure was the single movement area inspection before the first scheduled arrival, required by Aerodrome Manual 6.3.

That did not mean the hazard was unknown at the aerodrome. Its wildlife sighting log for 01/2025 to 06/2026 held 46 entries, and 41 of these recorded kangaroos on or near the movement area. Of those 41, 38 were timed between one hour before and two hours after last light. Eleven entries recorded animals on the sealed surface of runway 09/27. The log was a paper register kept in the aerodrome office, and it was not reported to any operator. The aerodrome reporting officer confirmed that no airline had ever asked for it.

The works register shows the same pattern for the physical barrier. Item WR-221, raised in 04/2023, recorded failed fence panels on the northern boundary between the 700 m and 1,300 m runway markers. It noted that animals had been seen passing through, and it estimated the cost of replacement at \$41,000. The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Item WR-263, raised in 02/2026, then recorded the same section as deteriorating further. When the site was inspected on 23/06/2026, a continuous 140 m section had the bottom two wires separated from the posts, with an animal pad through it at the 1,050 m mark. The kangaroos involved in the occurrence came onto runway 09 from the northern side, past the 1,100 m point.

The case for influence rests on what a documented plan would have provided. A plan is the mechanism that would have taken the aerodrome's sighting log to the operators using the aerodrome. It is also the mechanism that would have moved the failed fence from an unfunded works register onto a funded programme. A documented plan brings an audit trail and a named owner, and each of the other controls examined in this occurrence failed for want of exactly those two things.

The evidence against influence was given proper weight. Section 10.4.1 still required the aerodrome operator to take reasonable measures, and a documented plan is not the only way to repair a fence or to pass a sighting log to an airline. Moreover, the decisions to defer the fence repair were budget decisions taken by the council. There is no evidence that a documented plan would have attracted

funding where the works register had not. A plan is a document, whereas the fence repair required money.

The investigation resolved this on the balance of the evidence. The objection that a plan does not pay for a fence is correct as far as it goes. However, without a plan, nobody held a documented responsibility for bringing the fence, the sighting pattern and the needs of the airline together. In the absence of that responsibility, the deferral of the repair and the failure to share the sightings went unchallenged. On this basis it is likely that the threshold in AS-139 influenced the occurrence.

That assessment is less certain than the other influence findings in this analysis, and it should be read as such. The counter-argument was not disproved. It was outweighed.

The factor acted through two paths, and each is argued in its own subsection. The first is the deferral of the fence repair for three budget cycles, which was found very likely to have contributed and is analysed under Organisational Influences. The second is the operator's requirement for runway inspection only at listed aerodromes, where the listing criterion is a strike. This too was found very likely to have contributed, and it is analysed under Risk Controls.

The size threshold in AS-139 10.4.2, which left Tarlton Springs without a documented, owned and audited wildlife programme, is placed among the causal factors, with its influence assessed as likely rather than settled.

Organisational Influences

Perimeter fence repair deferred for three budget cycles

The aerodrome operator's works register shows that the northern perimeter fence had failed more than three years before the occurrence, and that the failure was known.

- Item WR-221, raised in 04/2023, described four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers. It noted that animals had been seen passing through and recommended replacement at an estimated \$41,000.
- The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. On each occasion it was put back for review at the next cycle.
- Item WR-263, raised in 02/2026, recorded the same section as deteriorating further.

None of this had to be reconstructed, because the aerodrome operator had set down both the condition and its deferral in its own register.

The site inspection on 23/06/2026 found what that history had produced. Along a continuous 140 m section of the northern fence, the bottom two wires had separated from the posts, and a clear animal pad ran through the fence at the 1,050 m mark (Figure 1). The runway survey recovered the two carcasses at 1,188 m and 1,221 m from the runway 09 threshold. The breach and both carcasses therefore lay on the northern side of the runway, with the pad a short distance before the strike position. This is consistent with the CVR, on which the first officer called the animals "on the left". It is also consistent with the occurrence notification, which records the kangaroos "moving onto the runway from the northern side, past the 1,100 m point."



Figure 1: The northern perimeter fence at the 1,050 m mark, photographed on the site visit: the fallen section and the animal pad through it. Site inspection photograph SV-2026-0417-07, northern boundary at the 1,050 m runway marker, taken 23/06/2026

The aerodrome's wildlife sighting log for 01/2025 to 06/2026 shows the exposure that the fence was there to contain. Of its 46 entries, 41 recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and eleven entries recorded animals on the sealed surface of runway 09/27. The log was a paper register held in the aerodrome office, and it was not reported to any operator.

A fence in the condition found does not exclude kangaroos. The pad through the breach, the side from which the animals were seen to come and the positions of the carcasses all point to the same section of boundary. It was therefore very likely that the deferred repair gave the two kangaroos their route onto the runway strip. The link from the AS-139 size threshold to this deferral was set out under External Influences and is not repeated here. The evidence places the deferral among the causal factors, because it provided the physical route without which the animals would not have reached the runway in this way.

The deferral of the northern fence repair was a causal factor, which very likely allowed the kangaroos onto the runway strip.

Evening service approved as a schedule change, route risk assessment not reopened

Route Risk Assessment RRA-041 issue 3, dated 11/2022, was the current assessment for Tarlton Springs. The hazards it assessed included "runway length and accelerate-stop margin" and "single runway with no lighting redundancy." It also carried the operating restriction "daylight operations only pending review of night lighting and diversion fuel policy."

Change Management Record CMR-2025-118 introduced an evening rotation from 14/07/2025, which was the first scheduled company operation into Tarlton Springs after last light. The record assessed the change as "schedule change only" and stated "No new hazard identified." To the question of whether a route risk assessment review was required, it answered "No." The Head of Flying Operations approved it on 22/05/2025.

CMR-2025-118 does not reference the daylight-only restriction anywhere. Nor was RRA-041 reopened, and the document register holds no issue 4. The Safety Department's records show no involvement in the change beyond receipt of the approved record. These are controlled documents, and nothing in the evidence contradicts them. It is virtually certain that the evening service was approved without the route risk assessment being reviewed.

The change process asked whether the change introduced something new. It did not ask whether the existing assessment still held for the new operation. Because the question was framed in that way, a sector that contradicted a live restriction passed as routine.

Why the change was assessed in this way could not be established. The approver has since left the company and could not be interviewed, so the report draws no conclusion about what was considered at the time of approval.

The influence of the non-review on the occurrence is less certain than the fact of it. A review of RRA-041 against a night operation would have brought two things into the same assessment: the aerodrome's wildlife exposure, and its absence from the Appendix 8A list of wildlife-active aerodromes. Two considerations weigh against that conclusion.

- The restriction names night lighting and diversion fuel only, so a review might have addressed those matters alone.
- Because the approver could not be interviewed, it is not known what information was in front of him.

Taking both into account, it is likely that reopening RRA-041 would have reached the wildlife hazard. The inspection control at listed aerodromes, and the absence of a runway inspection before this departure, are argued under Risk Controls and Individual Actions. The evidence places this approval among the causal factors, because it was the step at which the first after-dark operation began without the existing restriction being reassessed.

The approval of the evening service as a schedule change, without reopening RRA-041, was a causal factor, which likely meant the wildlife hazard at Tarlton Springs after last light was never assessed.

Risk Controls

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator had a defence written for exactly this hazard. Operations Manual Part A 8.14.2 states that "At an aerodrome listed in Appendix 8A, the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed." At every other aerodrome, 8.14.3 provides that "no specific action is required." Both the manual text and the appendix are documentary, so it is virtually certain that the arrangement existed as described.

The weakness lay in how an aerodrome came to be listed. Appendix 8A admits an aerodrome only where "a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months." No other evidence could bring an aerodrome onto the list, and nothing in Part A asks a crew to report a sighting. Tarlton Springs had never had a company strike, and so

it had never been listed. The list was last reviewed on 03/03/2025, four months before the evening service began.

The exposure was nonetheless known at the aerodrome itself. Its wildlife sighting log for 01/2025 to 06/2026 holds 46 entries, of which 41 record kangaroos on or adjacent to the movement area. Of those 41, 38 fall between one hour before and two hours after last light, and eleven record animals on the sealed surface of runway 09/27. The log is a paper register held in the aerodrome office and is not reported to any operator. The aerodrome reporting officer confirmed that no airline had ever asked to see it. Yet none of this could arm the operator's control, because a sighting is not a strike. At any aerodrome, the control could be triggered only by the outcome it existed to prevent.

Had the requirement applied, the crew would have requested an inspection, and they could not have begun the roll until the result was passed. The aerodrome's own staff disperse animals when they find them. It is therefore very likely that the requirement, had it applied, would have prevented the occurrence. The qualification is that an inspection completed some minutes before departure cannot guarantee a clear runway at the moment of the roll. The absence of an inspection request on the night is argued under Individual Actions. The organisational decisions that fed this criterion, namely the aerodrome size threshold and the approval of the evening service as a schedule change, are argued under External Influences and Organisational Influences.

The strike-based listing criterion was a causal factor, because it left the operator's runway inspection requirement unarmed at Tarlton Springs, which very likely allowed the take-off to begin with no inspection after last light.

Rejected take-off training covers aircraft system failures below V1 only

It is extremely likely that this gap existed as described. Module 4 of TRG-SYL-07 scripts four simulator scenarios: an engine failure at 90 kt, an engine fire at 100 kt, a configuration warning at 85 kt, and an engine failure at V1 plus 5 kt with the take-off continued. All four are aircraft system events, and three arise below V1. In the five training cycles held on file, no crew had been presented with an obstruction, an animal or any other external hazard on the runway. That includes the 214 crew who completed the module in the 2025 to 2026 cycle. Scenario records from before those five cycles were not available, and this is the only reason the finding stops short of virtual certainty.

Part B 4.6 matches the syllabus. It states that "At or above V1 the take-off shall be continued," and it contains no guidance on an obstruction seen on the runway at any speed. Each pilot, independently and without prompting, described every stop or go scenario they had been given as an engine event arriving below V1.

Whether the gap influenced the decision was argued, and both readings are set out here. The captain stated that he knew the rule and departed from it deliberately, having judged that continuing was the worse course. On that reading, training on obstructions would not have changed what he knew, and the gap would be a finding about the fleet rather than a cause of this reject.

The opposing reading starts from the time available. The decision came one second after the first officer's call, and it was made by a commander who had not seen the animals himself ("I never had them"). The first officer had them in sight for "two seconds. Maybe less." The captain was therefore considering the question for the first time. An unrehearsed decision made cold in that time is not the same decision as one made against a briefed and practised frame, and supplying that frame is the purpose of training. The investigation accepted this second reading as likely. It explains the circumstances of the decision and is not a judgement of it.

The absence of any external obstruction scenario from training and from Part B 4.6 was a causal factor, which likely shaped the decision to reject the take-off 6 kt above V1.

Cabin floor proximity emergency lighting partially unserviceable

This matter came to light through the first officer, who noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation. A functional check on 24/06/2026 armed the system on battery power, and two of the six floor proximity escape path segments on the left side aft of row 16 did not illuminate. The fault was traced to a failed ballast common to both segments, so it is virtually certain that the defect existed on the night.

The defect was latent. On this type, the pre-flight emergency lighting test illuminates the ceiling emergency lights and exit signs, but it neither illuminates nor monitors the floor segments. Without a dedicated functional test, neither the crew nor a line engineer could have detected the fault. The last such test had been carried out 11 months before the occurrence, within the maintenance programme's 24-month interval.

It is unlikely that the defect played any part in the occurrence or its outcome. No evacuation was commanded and no slides were deployed. Passengers left by the forward airstair with the main cabin lights on throughout, and the disembarkation was completed without urgency or incident.

It is very likely, however, that the defect matters beyond this event. A fire indication, smoke, or a night evacuation over the wing would each have placed weight on the aft escape path, and two dark segments there could separate an orderly exit from a blocked one. The deeper gap is in monitoring. The interval between functional tests is 24 months, the daily test cannot see the floor segments, and the same ballast is fitted across the type. As a result, no routine check establishes the condition of this defence elsewhere in the fleet.

The partially unserviceable floor proximity lighting did not contribute to the overrun or its outcome, but it was very likely a safety issue in its own right, because an escape path defence on a fleet-common component could fail undetected between functional tests.

Local Conditions

Departure in the hour after last light, the peak of kangaroo movement

The take-off roll began at 17:30 local. That was 32 minutes after sunset at 16:58, and 5 minutes after the end of civil twilight at 17:25. Moonrise was not until 22:41, and the moon was 14% illuminated. There was no natural ambient light on the strip, and no town lighting was within sight. The investigator's site visit at the same time of day on 23/06/2026 confirmed that the unlit graded strip on either side of the sealed surface cannot be seen from the centreline.

The aerodrome's wildlife sighting log places 38 of its 41 kangaroo entries between one hour before and two hours after last light, and eleven entries record animals on the sealed surface of runway 09/27. The astronomical data are documentary. The log, however, is a paper register whose completeness cannot be verified. For that reason it is extremely likely, rather than certain, that the departure took place within the aerodrome's period of peak kangaroo movement.

The light condition governed what the crew could see. After the aircraft stopped, the first officer said on the CVR that the animals "came out of the dark on the left of the strip", and that "There was nothing to see and then there were two of them." The animals were not detected until they were on the lit surface, which is what a dark strip would lead one to expect. The timing also coincides with the recorded movement peak. It is therefore likely that the departure in the hour after last light influenced the occurrence.

On its own, the condition is unremarkable. Its weight lies in the fact that it is what made the absence of a runway inspection matter, and that control is considered under Risk Controls. The evidence places this factor among the causal factors as the condition that brought the animals onto the strip at the time of the departure and kept them out of sight until they reached the sealed surface.

Runway edge lighting intensity reduced the crew's ability to detect the animals

The investigation examined whether the runway edge lighting made the animals harder to see, but the examination could not settle the question.

The intensity deviation itself is not in doubt. A photometric check on 25/06/2026 found four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal for stage 3, which is the section the animals crossed. The remaining lights were within tolerance. The aerodrome's maintenance programme checks lamp serviceability, but it does not measure intensity. In support of an effect on detection, the captain recalled at interview that the lighting was brighter than he would have wanted. He described a corridor of light with everything outside it black, from which anything off the seal could not be seen coming.

Against this, the first officer was the crew member who actually saw the animals. She described the lighting as normal and said that she saw them clearly once they were on the sealed surface, at around 200 m. She also said that, with only two daylight sectors into the aerodrome to compare against, she was not well placed to judge whether the lighting was unusual. Moreover, the captain's failure to see the animals has an explanation that does not depend on the lighting. He said on the CVR, "I never had them. I was heads-in on the engine page after V one", which is where the procedure places him.

Three things prevented the matter from being resolved. First, the only measurement was taken seven days after the occurrence, and nothing records what the lights were doing on the night. Second, the two accounts of the lighting do not agree. Third, no study links an edge light at 131% of nominal to the detection of an object on the sealed surface. The measurement shows a deviation, therefore, but not an effect. Resolving the matter would need evidence of the lighting output on the night, together with evidence connecting that output to detection, and neither is available. The investigation considered it about as likely as not that the lighting reduced the crew's ability to detect the animals. The matter remains unresolved, and nothing from it is carried into the conclusions.

Take-off commenced with a tailwind above limits

A tailwind was considered because the stop ran 62 m past the runway end, and it was found very unlikely to have existed. The aerodrome observations ("METAR YTLS 180700Z 09006KT" among them) and the recorded aircraft data both gave a headwind component throughout the roll, consistent with the 5 kt headwind used in the crew's performance entry.

Fifth sector, 11 hours into duty, 40 minutes behind schedule

The crew signed on at 06:30 local for five sectors. The fifth had been added that morning, when the rostered crew reported unfit. The occurrence came 11 hours 00 minutes into a duty planned at 12 hours 05 minutes, against a limit of 12 hours 30 minutes, and the aircraft departed 40 minutes late. The roster and duty record are documentary, so it is virtually certain that these conditions existed.

A late fifth sector at that point in a long duty would ordinarily invite the question of degraded decision making. Both crew described themselves at interview as tired but not struggling. More to the point, the recorded data contradict each mechanism through which fatigue would have had to act. The FDR shows 0.6 seconds from decision to power levers at flight idle and 2 seconds to

maximum braking, at the fast end of the operator's fleet distribution, and there was no error of omission on the roll. The one decision that the schedule might plausibly have affected was not requesting a runway inspection. Yet no document prompted either crew member to consider that step, so fatigue cannot account for its omission. It is therefore unlikely that the duty conditions had any influence on the occurrence.

The exposure nonetheless remains. The evening Tarlton Springs rotation routinely places the last sector in the final hour of a 12-hour duty. It does so at an aerodrome with no company engineering, no alternate within 200 nm at night and no crewing resilience. A single unfit crew member turns a four-sector day into a five-sector one on the morning of operation. It is likely that this is a matter worth acting on, and it is reported as a finding about the schedule rather than about this occurrence.

Individual Actions

No runway inspection requested before the first departure after last light

Neither crew member requested a runway inspection before the take-off, and the tower did not offer one. This is virtually certain, because two independent recordings and the aerodrome's own log agree. Neither the CVR nor the tower's recorded audio contains a request for an inspection, an offer of one, or any mention of wildlife before departure. The aerodrome's daily inspection log for 18/06/2026 shows a single inspection at 07:35 local, before the first arrival, with the result "movement area clear". No later inspection was carried out or requested.

The inspection was available. Aerodrome Manual 6.3 provides for an inspection outside the scheduled time at the request of a pilot in command, and the reporting officer confirmed that one would have been carried out if asked. Both flight crew stated that they did not consider it. The captain had expected Tarlton Springs to appear on the Appendix 8A list and was surprised to find that it did not. Because the aerodrome was not listed, Part A 8.14.3 required no specific action, and nothing in the crew's documentation prompted the request. The omission therefore had no owner. It arose from the structure of the operator's control, which is argued under Risk Controls, and not from a lapse by either pilot.

An inspection would very probably have mattered to the outcome. It would have covered the 1,050 m to 1,250 m section of runway 09, where the fence pad and both carcasses were later found, and aerodrome staff disperse animals when they find them. Against this, an inspection made some minutes before the roll cannot guarantee that the runway stays clear, since animals may enter the strip after the vehicle has passed. For that reason the influence of the omission is put at likely rather than higher.

The absence of a runway inspection was likely a causal factor, because without it the animals were not looked for or dispersed before the roll began.

Take-off rejected 6 kt above V1

The FDR shows V1 at 112 kt at 07:30:29, and both power levers retarded to flight idle from 07:30:32 at 118 kt, three seconds and six knots after V1. The CVR shows that the decision was made and announced in a single exchange. The first officer called "Roos - roos, roos, on the left" at 07:30:31, and one second later the captain responded "Stopping. I have control." The existence of the late reject is virtually certain.

Part B 4.6.4 states that "At or above V1 the take-off shall be continued." Section 4.6 contains no guidance on an obstruction seen on the runway at any speed, and it does not address a case in which continuing is itself hazardous. In interview and in his air safety report, the captain explained his judgement. Rotating into two large animals would have left him airborne at night with damage

he could not assess, over country offering no usable option. He chose the risk whose end he could see, and he acknowledged that this was not by the book.

The consequence for stopping distance was direct. The scheduled accelerate-stop distance of 1,780 m assumes that the reject begins at or before V1, and the margin between the distance required and the distance available was 67 m. At the moment of the reject, the aircraft was 1,152 m along runway 09, with 695 m of sealed surface remaining. The stop actually achieved from 118 kt took 757 m. Beginning the stop six knots and three seconds late therefore consumed the 67 m margin and more, even before the loss of No. 2 reverse is taken into account. It is extremely likely that the late reject contributed to the overrun.

The decision sits downstream of other factors. The animals' presence on the sealed surface is a matter for Occurrence Events. The scope of rejected take-off training, which covers aircraft system failures below V1 only, is argued under Risk Controls. Whether the runway edge lighting affected detection could not be resolved, and it is discussed under Local Conditions.

The reject above V1 was extremely likely a causal factor, because it committed the aircraft to a stop for which the remaining runway was insufficient.

Maximum braking and No. 1 reverse applied without delay

Once the decision was made, the stop was executed quickly. The FDR shows the power levers at flight idle within 0.6 seconds, and maximum braking established at 07:30:34, 2 seconds after the decision. By comparison, the operator's recorded fleet mean from decision to maximum braking is 3.2 seconds. Anti-skid was active on all four wheels throughout, and brake energy reached 84% of the certified maximum without a brake fire. No. 1 reverse was selected at 07:30:36, 4 seconds after the decision. No. 2 reverse was unavailable, because its propeller had been damaged in the strike. The first officer made the speed callouts and the call that the runway end was approaching.

The aircraft left the sealed surface at 21 kt and stopped 62 m into the 90 m RESA, upright and on the extended centreline. It was estimated that a stop begun at the fleet mean response would have carried the aircraft a further 30 to 40 m, close to the far edge of the RESA. The overrun was therefore a low-speed excursion onto a prepared surface, rather than a departure from the RESA at speed. The finding that the reject fell outside procedure does not diminish this, since the stop that followed it was executed faster than the fleet average.

The prompt application of maximum braking and No. 1 reverse very likely reduced the severity of the overrun.

Occurrence Events

Two kangaroos entered the runway strip and moved onto the sealed surface

It is virtually certain that two kangaroos were on the sealed surface of runway 09 during the take-off roll, and three separate sources each establish this on their own. The runway survey recovered two adult kangaroo carcasses on the northern half of runway 09, at 1,188 m and 1,221 m from the threshold. The CVR records the first officer's call at 07:30:31, "roos, roos, on the left". The FDR records an impact at 07:30:33.

The first officer saw the animals crossing from the northern side of the runway towards the south, and estimated their range at first sight as around 200 m. After the aircraft had stopped, the first officer told the commander that they had come "out of the dark on the left of the strip" and had been in view for "two seconds. Maybe less."

The positions of the carcasses are consistent with the animals having come from the north. The animal pad through the northern perimeter fence lies at the 1,050 m mark, which is 138 m before

the first carcass and on the same side of the runway. The route by which the animals reached the strip is, however, a separate question from whether they reached the sealed surface, and it is argued with the deferred repair of the fence under Organisational Influences.

Kangaroos on the runway were not an isolated event at this aerodrome. The aerodrome's wildlife sighting log for 01/2025 to 06/2026 holds 46 entries. Of these, 41 record kangaroos on or adjacent to the movement area, and eleven record animals on the sealed surface of runway 09/27. No runway inspection had been carried out since the one that morning.

The animals' presence led directly to both the rejected take-off and the strike. Up to the reject, the recorded data shows no other cue. No engine caution, warning or exceedance was recorded, and the CVR contains no other call. The commander's "Stopping. I have control." came one second after the sighting call. Without the animals on the surface there would have been neither a reject nor a strike, so their presence is a causal factor.

It is therefore concluded that two kangaroos entering the runway strip and moving onto the sealed surface extremely likely led to both the rejected take-off and the strike, and that this was a causal factor in the overrun.

No. 2 propeller and right main gear struck two kangaroos

The strike is virtually certain, because the recorded data, the physical damage and the recovered tissue all agree. The FDR records the impact at 07:30:33 at 114 kt, one second after the reject began. It shows a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g.

The post-occurrence inspection found damage consistent with the strike:

- Three No. 2 propeller blades were damaged beyond repairable limits, and one blade tip had separated.
- Animal tissue was recovered from two blade roots.
- The right main gear doors were distorted.
- The lower right fuselage skin was punctured by propeller debris.

The strike had a direct and measurable effect on the stop. No. 2 propeller RPM decays from the moment of impact, and the No. 2 reverse interlock did not make. Reverse was therefore selected on No. 1 only, at 07:30:36, and the CVR records the commander's remark that "number two's not coming". Because the scheduled accelerate-stop figures assume symmetric reverse, the strike removed half of the available reverse thrust early in a stop that had only a 67 m scheduled margin to begin with.

It is concluded that the strike virtually certainly reduced the deceleration achieved, and that it was a causal factor in the overrun.

Deceleration achieved below the scheduled accelerate-stop performance

The scheduled accelerate-stop distance required was 1,780 m against an ASDA of 1,847 m, a margin of 67 m. That figure assumed that the stop would begin at or before V1 and would use symmetric reverse. Neither assumption held on this occasion.

Distance reconstruction from recorded groundspeed places the aircraft 1,152 m along runway 09 at the reject, leaving 695 m of sealed surface. The stopping distance actually achieved from 118 kt was 757 m. The difference of 62 m matches the aircraft's final position, 62 m into the 90 m RESA. Both the continuous tyre marks from 1,160 m to the runway end and the nose wheel rut on the extended centreline agree with this reconstruction.

Under maximum braking, the mean deceleration recorded was 2.6 m/s². This is approximately 15% below the value used in the certified data for this weight and runway condition. The aircraft left the sealed surface at about 21 kt, after which deceleration fell to 1.9 m/s² in the graded gravel.

The evidence does not suggest that the shortfall came from the brakes. Maximum braking was established 2 seconds after the decision, and anti-skid was active on all four wheels. The anti-skid system was found serviceable after the occurrence, and the brakes absorbed 84% of the certified maximum energy for the weight and speed.

Two separate routes lead to this shortfall, but it is one mechanism. The late start of the stop is argued under Individual Actions, and the loss of No. 2 reverse is argued above. The investigation assessed that both were needed to carry the aircraft beyond the runway end, and that neither alone would have done so. The overrun is the shortfall expressed as a distance.

It is concluded that the deceleration shortfall, measured against a stop begun 6 kt above V₁, virtually certainly produced the overrun, and that it was a causal factor.

No. 2 engine power loss during the take-off roll

A No. 2 engine power loss was considered as a cause of the reject and was found very unlikely. Both engines held take-off torque within 0.1% of each other until the power levers were retarded at 07:30:32, and no caution, warning or engine call was recorded before that point. The borescope found no pre-existing damage, and the No. 2 RPM decay began only at the impact at 07:30:33, one second after the reject.

Conclusions

The investigation identified the following causal and contributory factors. They should not be read as apportioning blame or liability to any organisation or individual.

A finding stated without a likelihood term was established as fact.

Findings

1. Aerodrome Standard AS-139 section 10.4.2 required a documented wildlife hazard management plan only where an aerodrome was used by aeroplanes above 27,000 kg MTOW or served more than 350,000 passenger movements a year. (Safety issue)
2. Tarlton Springs met neither limb of that threshold, with a heaviest scheduled aeroplane of 22,900 kg and 61,400 passenger movements in 2025. It held no documented plan, hazard assessment or dispersal programme, and it was not in breach of the standard.
3. The failure of the northern perimeter fence was recorded in works register item WR-221 in 04/2023, with animals observed passing through, and its repair was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. (Safety issue)
4. The site inspection on 23/06/2026 found a continuous 140 m section of the northern fence with the bottom two wires separated from the posts and an animal pad through it at the 1,050 m mark, on the same side of the runway as both carcasses.
5. Route risk assessment RRA-041 issue 3 restricted Tarlton Springs to daylight operations pending review of night lighting and diversion fuel policy.
6. Change management record CMR-2025-118, approved on 22/05/2025, introduced the evening Tarlton Springs rotation from 14/07/2025 as a schedule change only. It answered "No" to whether a route risk assessment review was required, and RRA-041 was not reopened. (Safety issue)
7. Operations Manual Part A 8.14.2 required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A, and the only route onto that list was a wildlife strike involving a company aircraft reported at the aerodrome in the preceding 24 months. (Safety issue)
8. Tarlton Springs had never been listed in Appendix 8A, and no provision of Part A asked crews to report a sighting or admitted an aerodrome to the list on any other evidence.
9. The aerodrome's wildlife sighting log held 41 kangaroo entries. Of these, 38 were around last light and eleven recorded animals on the sealed surface. The log had never been passed to or requested by any operator.
10. The four scripted rejected take-off scenarios in TRG-SYL-07 module 4 were all aircraft system events, and none of the five training cycles on file presented an external obstruction scenario. Part B 4.6 contained no guidance on an obstruction on the runway at any speed. (Safety issue)
11. Two of the six cabin floor proximity escape path segments aft of row 16 on the left side were dark because of a failed common ballast. The pre-flight emergency lighting test did not illuminate or monitor these segments, and the last functional test had been 11 months earlier, within the 24-month programme interval. (Safety issue)
12. The take-off roll began 32 minutes after sunset and 5 minutes after the end of civil twilight, with no moon. This was within the period in which the aerodrome's own log recorded most of its kangaroo movement.

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13. The headwind component was 4 to 6 kt throughout the roll, and there was no tailwind component at any point.
14. The flight was the fifth sector of the crew's duty. It had been added that morning, it was 11 hours 0 minutes into a planned duty of 12 hours 5 minutes against a limit of 12 hours 30 minutes, and it departed 40 minutes behind schedule.
15. No runway inspection was requested or offered before departure, and the only inspection that day had been made at 07:35 local, before the first arrival. The aerodrome reporting officer confirmed that an inspection would have been carried out if asked.
16. The pilot in command rejected the take-off at 118 kt at 07:30:32, 3 seconds and 6 kt after V1 was called at 112 kt. This was contrary to Part B 4.6.4, and he described it as a deliberate decision.
17. Both engines were symmetric and free of any caution, warning or exceedance until the reject, and the No. 2 propeller RPM decay began at the moment of impact.
18. The No. 2 propeller and right main gear struck the two kangaroos at 07:30:33 at 114 kt. The strike damaged three No. 2 propeller blades beyond repairable limits, and the No. 2 reverse interlock did not make.
19. The power levers reached flight idle 0.6 seconds after the decision, and maximum braking was established 2 seconds after it, against a recorded fleet mean of 3.2 seconds.
20. The scheduled accelerate-stop distance of 1,780 m against an ASDA of 1,847 m assumed a stop begun at or before V1 with symmetric reverse. The stop actually achieved from 118 kt was 757 m against 695 m of sealed surface remaining, with a mean deceleration approximately 15% below the certified value.
21. The aircraft left the end of runway 09 at about 21 kt and came to rest 62 m into the 90 m RESA, upright on the extended centreline and 28 m short of the end of the prepared area.

Causal factors

- It is virtually certain that a documented wildlife plan was required only above a size threshold the aerodrome did not meet. It is likely that this contributed to the occurrence, although the point was argued rather than settled, because the fence deferrals were budget decisions that a documented plan might not have changed. (Safety issue)
- It is virtually certain that the perimeter fence repair was deferred for three budget cycles, and very likely that this contributed to the occurrence by leaving open the route through which the animals reached the runway. (Safety issue)
- The departure very likely took place in the hour after last light, the peak of kangaroo movement at the aerodrome, and this likely contributed to the occurrence because the animals could not be seen until they were on the lit surface.
- With virtual certainty, the evening service was approved as a schedule change and the route risk assessment was not reopened. This likely contributed to the occurrence, although a review of RRA-041 might have addressed only the lighting and fuel matters its restriction named, and the approver could not be asked what was before him. (Safety issue)
- Runway inspection was required only at listed aerodromes and the listing criterion was a strike, which is virtually certain, and this very likely contributed to the occurrence because Tarlton Springs had no reported strike and was never listed. (Safety issue)
- It is virtually certain that no runway inspection was requested before the first departure after last light. It is likely that this contributed to the occurrence, the qualification

reflecting that an inspection some minutes before the roll could not have guaranteed a clear runway.

- That two kangaroos entered the runway strip and moved onto the sealed surface is virtually certain, and it is very likely that this contributed to the occurrence, since both the reject and the strike followed from their presence.
- It is very likely that rejected take-off training covered aircraft system failures below V1 only. It is likely that this contributed to the occurrence, although the pilot in command knew the rule and judged continuing to be the worse risk, so the gap may explain the decision without having changed it. (Safety issue)
- The take-off was rejected 6 kt above V1, which is virtually certain, and this very likely contributed to the occurrence because a stop begun 3 seconds late consumed the 67 m scheduled margin.
- It is virtually certain both that the No. 2 propeller and right main gear struck two kangaroos and that this contributed to the occurrence, by removing the No. 2 reverse early in the stop.
- Deceleration achieved below the scheduled accelerate-stop performance was virtually certain and virtually certainly contributed to the occurrence, since the 757 m stop from 118 kt exceeded the 695 m of sealed surface remaining by 62 m.

Contributory factors

- It is virtually certain that the cabin floor proximity emergency lighting was partially unserviceable, and very likely that this increased risk to occupants in any evacuation. It did not contribute to the overrun or its outcome, because no evacuation was commanded and the main cabin lights were on throughout. (Safety issue)
- The flight was the fifth sector, 11 hours into duty and 40 minutes behind schedule, which is virtually certain, and this likely increased risk as a standing exposure on the evening Tarlton Springs schedule. The recorded data contradicted each mechanism by which it could have contributed to this occurrence.

Other findings

- The hypothesis that the take-off commenced with a tailwind above limits was examined and ruled out. It is very unlikely that any tailwind was present, because the aerodrome observations and the recorded aircraft data both showed a headwind component throughout the roll.
- A No. 2 engine power loss during the take-off roll was also examined and ruled out. It is very unlikely that any loss occurred before the reject, since both engines remained within 0.1% torque of each other until 07:30:32 and the No. 2 RPM decay began at impact at 07:30:33.
- Maximum braking and No. 1 reverse were applied without delay, with maximum braking established 2 seconds after the decision against a fleet mean of 3.2 seconds. This very likely reduced the severity of the outcome, because a stop at the fleet mean response was estimated to have carried the aircraft a further 30 to 40 m, close to the far edge of the RESA.

The investigation also examined whether runway edge lighting intensity reduced the crew's ability to detect the animals, and it could not settle the matter. It is therefore not a finding. A photometric check on 25/06/2026 found four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal for stage 3, in the section the animals crossed. The aerodrome's

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maintenance programme checked lamp serviceability but never measured intensity. The pilot in command recalled the lighting as brighter than he would want, with everything outside it dark. Against this, the first officer, who was the crew member who saw the animals, described the lighting as normal and saw them clearly once they were on the sealed surface. She had only daylight landings at the aerodrome for comparison. The pilot in command's own non-detection is explained without reference to the lighting, because after V1 his attention was on the engine page, where the procedure placed it. The intensity deviation itself is established. The step from that reading to a reduced ability to see an animal on the runway could not be taken, for three reasons: the check was made seven days after the occurrence, nothing recorded what the lights were doing on the night, and no study links an edge light at that intensity to the detection of an object on the runway. The evidence was evenly balanced. Resolving it would require a measurement of the lights' output on the night, which does not exist, together with evidence linking edge light intensity to detection.

Safety action and recommendations

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Aerodrome Standard AS-139 requires a documented wildlife hazard management plan only where an aerodrome is used by aeroplanes above 27,000 kg MTOW, or where it serves more than 350,000 passenger movements a year. Tarlton Springs met neither condition. Its heaviest scheduled aeroplane was 22,900 kg, and it handled 61,400 passenger movements in 2025. Its wildlife measures therefore did not need to be documented. The aerodrome held no plan, no hazard assessment and no dispersal programme. It was not in breach of the standard.

The issue matters because the threshold in the standard left an aerodrome of this size with a wildlife programme that was undocumented, unaudited and, in practice, unfunded. It is likely that the absence of a documented plan contributed to the occurrence. Such a plan would have given the programme an owner and an audit trail. It would also have been the means of passing the aerodrome's sighting log to the operators using it, and of placing the failed fence on a funded programme.

That conclusion is held with less confidence than the other findings on contribution in this report. The aerodrome operator remained obliged to take reasonable measures whether or not a plan was documented. There was also no evidence that a documented plan would have been funded any better than the deferred fence repair. The issue led to two further factors: the deferral of the perimeter fence repair over three budget cycles, and a runway inspection requirement that applied only at aerodromes listed after a strike.

No safety action has been recorded in response to this issue, and no recommendation is made in respect of it.

Perimeter fence repair deferred for three budget cycles

The repair of the northern perimeter fence at Tarlton Springs was deferred for three budget cycles. Works register item WR-221 was raised in 04/2023. It recorded four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers, with animals observed passing through. Replacement was estimated at \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds, and each entry read "defer, unfunded, review next cycle". Item WR-263, raised in 02/2026, recorded the same section deteriorating further. On the site visit the investigation found a continuous 140 m section with the bottom two wires separated from the posts. An animal pad ran through the fence at the 1,050 m mark (Figure 1).

The significance of the issue lies in the route it left open. A fence in that condition does not exclude kangaroos. The pad through the breach lies upstream of the strike position, on the same side of the runway. The deferred repair very likely contributed to the two kangaroos entering the runway strip.

Safety action taken

The aerodrome operator funded item WR-221 from its 2026 contingency. It programmed replacement of the 140 m northern boundary fence for 09/2026, and it installed a temporary electric wire on 25/06/2026. This action was recorded as in progress. The permanent replacement has not been reported complete, so the action answers the issue in part.

Safety recommendations

No safety recommendation has been recorded for this issue.

Evening service approved as a schedule change, route risk assessment not reopened

The operator approved its evening service into Tarlton Springs as a schedule change and did not reopen the route risk assessment for the aerodrome. Route Risk Assessment RRA-041 issue 3 is the current issue. It restricted Tarlton Springs to "daylight operations only pending review of night lighting and diversion fuel policy". CMR-2025-118 introduced a scheduled sector after last light without referring to that restriction, and RRA-041 was neither reviewed nor reissued. The Safety Department's records show no involvement beyond receipt of the approved change record.

The safety significance lies in the way the change was assessed. The process asked whether the change was new rather than whether the existing assessment still held. As a result, a company aircraft operated into the aerodrome after last light for the first time, in direct contradiction of a live restriction. A review of RRA-041 against a night operation would likely have reached the aerodrome's wildlife exposure and the Appendix 8A listing in the same pass. It might, however, have addressed only the lighting and fuel matters that the restriction names. The reasons for the assessment could not be established, because the approver has left the company and could not be interviewed.

After the occurrence, the Head of Flying Operations suspended the evening Tarlton Springs service. No safety action has been recorded that addresses the change management process through which the service was approved, or the review of RRA-041. [NOT ESTABLISHED: whether the operator has taken or intends to take safety action on this issue, and whether the suspension of the evening service is to be treated as action against it.] No safety recommendation is made on this issue.

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

Operations Manual Part A 8.14.2 required the pilot in command to request a runway inspection before the first departure after last light at any aerodrome listed in Appendix 8A. The roll could not begin until the result was passed. At other aerodromes nothing was required. An aerodrome was listed only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. No provision asked crews to report a sighting, and none admitted an aerodrome to the list on any other evidence. Tarlton Springs had never been listed.

This was the operator's own defence against exactly this kind of occurrence, and its absence was found very likely to have contributed. The control itself was sound. Its trigger, however, could be met only by the event it existed to prevent. The aerodrome's wildlife sighting log held eleven entries of kangaroos on the sealed surface. None of them could bring the aerodrome onto the list, because they were not strikes and had not been reported to the company.

In response, the Head of Flying Operations issued temporary revision TR-2026-04 to Appendix 8A on 26/06/2026, listing Tarlton Springs. The inspection requirement now applies to every departure there after last light, pending a permanent change. This action answers the issue only in part. It arms the control at one aerodrome but leaves the listing criterion unchanged everywhere else.

It is therefore recommended that the operator replace the Appendix 8A listing criterion so that an aerodrome is listed on its wildlife exposure rather than on a reported strike. It is further recommended that the operator add a route risk assessment trigger for any aerodrome first operated into after last light.

Rejected take-off training covers aircraft system failures below V1 only

The operator's recurrent training syllabus, TRG-SYL-07, presented rejected take-off decisions only as aircraft system events. Module 4 contained four scripted scenarios: an engine failure at 90 kt, an engine fire at 100 kt, a configuration warning at 85 kt, and an engine failure at V1 plus 5 kt in which the take-off was continued. No crew had been given an obstruction on the runway in any of the five training cycles on file. The current cycle alone covered 214 crew. Operations Manual Part B section 4.6 matched the syllabus. It listed the reject items to V1 and required the take-off to be continued at or above V1, but it gave no guidance on an obstruction on the runway at any speed. It is extremely likely that this gap existed. The only limit on that assessment is that scenario records beyond five cycles were not available.

The gap matters because the captain had to weigh an obstruction above V1 for the first time, in two seconds, with no briefed or practised frame to decide against. It is likely that this contributed to the take-off being rejected 6 kt above V1. That finding is qualified. The captain knew the rule and departed from it deliberately, because he judged continuing to be the worse course. On that reading the gap is a finding about the preparation of the fleet's crews rather than the reason for this reject. The training scope explains the conditions in which the decision was made. It does not change what the captain knew.

No safety action has been recorded in response to this safety issue, and no recommendation is made.

Cabin floor proximity emergency lighting partially unserviceable

The cabin floor proximity emergency lighting on VH-SBQ was found to be partially unserviceable, and it is virtually certain that this condition existed at the time of the occurrence. While walking back through the cabin during the disembarkation, the first officer noticed that sections of the aisle floor strip were unlit. A functional check on 24 June 2026 then found two of the six floor proximity escape path segments aft of row 16 on the left side dark on battery power with the system armed. The defect was traced to a failed common ballast.

The defect had no bearing on the overrun or its outcome. No evacuation was commanded and the main cabin lights were on throughout. The disembarkation by the forward airstair took 4 minutes 8 seconds without incident. The defect is reported because it very likely increased risk. A partially dark escape path is a defence that is absent when it is needed. In a fire, in smoke, or in a night evacuation over the wing, two dark segments in the aft cabin would be the difference between an orderly exit and a blocked one.

Neither the crew nor a line engineer could have detected the defect. The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs, but it neither illuminates nor monitors the floor segments. Only a dedicated functional test would reveal the defect. The last such test had been carried out 11 months before the occurrence, within the 24-month programme interval. The same ballast is fitted across the type, so the condition of the floor proximity lighting across the fleet is not known.

No safety action has been recorded in response to this safety issue.

Distribution

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Fictional case study for training. Not a real investigation.

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